

## Experiment No. 8

### Experiment Name

### Application of Optical Flow for Real-Time Object Tracking and Motion Analysis

#### Aim

To implement an optical flow-based motion analysis system for real-time object tracking and evaluate its effectiveness in detecting and tracking moving objects in video sequences using Python and OpenCV.

#### Purpose

While optical flow estimates the motion between consecutive video frames, its true potential lies in real-world applications such as object tracking, traffic monitoring, human activity recognition, autonomous navigation, video surveillance, and sports analytics. By analyzing motion vectors, computer vision systems can continuously track moving objects without relying solely on object detection algorithms.

In this experiment, students will develop a real-time object tracking application using optical flow techniques and analyze object trajectories, motion direction, and movement patterns in video sequences.

#### Prerequisites

- Basic knowledge of Digital Image Processing
- Familiarity with Python programming
- Python libraries: OpenCV (cv2), NumPy, and Matplotlib
- Understanding of Optical Flow algorithms (Lucas-Kanade and Farneback)
- Jupyter Notebook, Google Colab, or Visual Studio Code

#### Steps

1. Import the required libraries and load a real-world video sequence or capture live video using a webcam.
2. Identify a moving object or select feature points using the Shi-Tomasi Corner Detection algorithm.
3. Apply the Lucas-Kanade Optical Flow algorithm to track the movement of the selected object across consecutive video frames.
4. Draw motion trajectories and displacement vectors to visualize the path followed by the moving object.

5. Calculate the direction and magnitude of object motion and display the tracking results in real time.
6. Repeat the experiment using the Farneback Dense Optical Flow algorithm and compare the tracking performance.
7. Evaluate the impact of object speed, illumination changes, occlusion, and camera movement on tracking accuracy.
8. Analyze the tracked motion to estimate parameters such as object displacement, trajectory, and average speed.
9. Compare the effectiveness of optical flow-based tracking with conventional frame-by-frame object detection techniques.
10. Document the observations and discuss practical applications of optical flow in intelligent vision systems.

### Questions

1. How does object tracking differ from object detection?
2. Explain how Optical Flow can be used for real-time object tracking.
3. What is the role of Shi-Tomasi Corner Detection in the Lucas-Kanade Optical Flow algorithm?
4. Why is Optical Flow suitable for motion analysis in videos?
5. What challenges arise while tracking fast-moving or partially occluded objects?
6. Compare Optical Flow-based tracking with deep learning-based object tracking methods.
7. How can Optical Flow be used in traffic monitoring and autonomous driving systems?
8. Explain the effect of camera motion on Optical Flow estimation.
9. Mention five real-world applications where motion analysis using Optical Flow is commonly employed.
10. How can Optical Flow improve the performance of surveillance, robotics, and human activity recognition systems?